Algebraic Numbers and Algebraic Integers

Given a field K of characteristic 0, the natural homomorphism $\mathbb{Z} \to K$ is one-to-one. Hence, we identify $\mathbb{Z}$ as a subring of K. In addition, we can also think of $\mathbb{Q}$ as a subfield of K and thus K is a vector space (possibly of infinite dimension) over $\mathbb{Q}$.

Exercise: Given elements $\alpha_1, \ldots, \alpha_n$ of K, note that they are linearly independent over $\mathbb{Q}$ if and only if the natural group homomorphism $\mathbb{Z}e_1 \oplus \cdots \oplus \mathbb{Z}e_n \to K$ given by $e_i \mapsto \alpha_i$ for all i, is one-to-one. (Hint: Given any relation $\sum_i q_i\alpha_i$ with $q_i \in \mathbb{Q}$, we can "clear denominators" to obtain a relation with integer coefficients.)

Given elements $\alpha_1, \ldots, \alpha_n$ of K, we say that they are *algebraically independent* over $\mathbb{Z}$ if the natural ring homomorphism $\mathbb{Z}[X_1, \ldots, X_n] \to K$ given by $X_i \mapsto \alpha_i$ for all i, is one-to-one. We say that these elements are *algebraically dependent* otherwise.

In particular, we can define the following notions for a single element α in K:

- an element α in K is called *transcendental* if, for every non-zero polynomial $P(X)$ with integer coefficients we have $P(\alpha) \neq 0$.
- an element α in K is called *algebraic* if, there is a non-zero polynomial $P(X)$ with integer coefficients such that $P(\alpha) = 0$.

Note that, in the latter case, the ring homomorphism $\mathbb{Z}[X] \to K$ given by $X \mapsto \alpha$ has a non-zero kernel. It follows that the ring homomorphism $\mathbb{Q}[X] \to K$ has a non-zero kernel. Since $\mathbb{Q}[X]$ is a principal ideal domain, this kernel is generated by a single irreducible polynomial $P(X)$. By clearing denominators, we can assume that this polynomial has integer coefficients and content 1. (Recall that the content of an integer polynomial is the greatest common divisor of its coefficients.)

Exercise: When α in K is algebraic show that the kernel of $\mathbb{Z}[X] \to K$ is generated by a single irreducible polynomial $P(X)$ with integer coefficients and content 1. If we assume that the leading coefficient of $P(X)$ is positive, then it is *uniquely* determined by α.

This polynomial is called the *minimal polynomial* of α. The degree of this polynomial is called the degree of α.

Note: In some texts a looser definition of minimal polynomial is given which is only determined upto a non-zero rational number. We will use the above stricter definition as we can then talk about *the* minimal polynomial.

An element α in K which is algebraic is also called an *algebraic number* when we think of K as consisting of "numbers" (for example, when K is one of the fields $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}_p$.).

Algebraic Integers. An element α of K is said to be *integral* if there is a non-zero *monic* polynomial $P(X)$ with integer coefficients such that $P(\alpha) = 0$.

When we think of K as consisting of "numbers", we also use the term *algebraic integer* for such a number α.

Exercise: Suppose that $P(X)$ and $Q(X)$ are in $\mathbb{Z}[X]$ such that $Q(X)$ has content 1, $P(X)$ is a monic polynomial, and $Q(X)$ divides $P(X)$ in $\mathbb{Q}[X]$. Show that $Q(X)$ is monic.

It follows that an algebraic element $\alpha \in K$ is integral if and only if its minimal polynomial (as defined above) has leading term of the form X^d where d is the degree of α.

Finite dimensional $\mathbb{Q}$-subspaces

When $\alpha \in K$ is algebraic, the image of $\mathbb{Q}[X]$ in K is a field $\mathbb{Q}[\alpha]$ which is spanned as a vector space over $\mathbb{Q}$ by $1, \alpha, \dots, \alpha^{d-1}$ where d is the degree of α.

Observe that $V = \mathbb{Q}[\alpha]$ is a finite dimensional $\mathbb{Q}$-subspace of K such that $\alpha \cdot V \subset V$.

Conversely, suppose that an element $\alpha \in K$ has the property that there is some non-zero finite dimensional $\mathbb{Q}$-subspace $V \subset K$ such that $\alpha \cdot V \subset V$. Using some $\mathbb{Q}$-basis $S = \{v_1, \dots, v_n\}$ of V over $\mathbb{Q}$ we can express multiplication by α as a matrix by writing

$$M_S(\alpha)(v_i) = \alpha \cdot v_i = \sum_{j=1}^{n} a_{i,j} v_j$$

where $a_{i,j}$ are rational numbers. In other words, $M_S(\alpha)$ is an $n \times n$ matrix over $\mathbb{Q}$. Since V is non-zero, we see that α is an *eigenvalue* of this matrix when we consider it as matrix with entries in K. It follows that α satisfies the characteristic polynomial $\det(XI_n - M_S(\alpha))$ of $M_S(\alpha)$. This is a polynomial with rational coefficients, so as usual we can clear denominators to obtain a polynomial with integer coefficients satisfied by α. Thus, we conclude

> An element α in K if algebraic if and only if there is a non-zero finite dimensional $\mathbb{Q}$-subspace V of K such that $\alpha \cdot V \subset V$.

Finitely generated subgroups

We now extend this idea to integral elements.

When $\alpha \in K$ is integral, the image $\mathbb{Z}[\alpha]$ of $\mathbb{Z}[X]$ in K is generated as a *group* by $1, \alpha, \dots, \alpha^{d-1}$ where d is the degree of α.

Observe that $H = \mathbb{Z}[\alpha]$ is a finitely generated subgroup of the additive group of K such that $\alpha \cdot H \subset H$.

Conversely, suppose that an element $\alpha \in K$ has the property that there is some non-zero finitely generated subgroup $H \subset K$ such that $\alpha \cdot H \subset H$. Then

$\mathbb{Q} \cdot H = V$ is a finite dimensional $\mathbb{Q}$-subspace of K; let $v_1, \ldots, v_n$ be a $\mathbb{Q}$-basis of V.

Exercise: There is a choice of $v_1, \ldots, v_n$ such that H is a subgroup of $\mathbb{Z}v_1 + \cdots + \mathbb{Z}v_n$. (Hint: For each element $v \in V$, there is a common denominator for all q_i in an expression of the form $v = \sum_i q_i v_i$ with q_i rational numbers. Since H is finitely generated, we can choose a common denominator for all its generators.)

Since $v_1, \ldots, v_n$ are linearly independent, the group $\mathbb{Z}v_1 + \cdots + \mathbb{Z}v_n$ is a free Abelian group. As H is a subgroup of a free Abelian group, it is also a free Abelian group. Moreover, since $\mathbb{Q} \cdot H = V$, we see that H has rank n. Thus, we can write $H = \mathbb{Z}h_1 + \cdots + \mathbb{Z}h_n$ where $U = \{h_1, \ldots, h_n\}$ *also* give a $\mathbb{Q}$-basis for $V = \mathbb{Q} \cdot H$.

As above, we can express multiplication by α as a matrix by writing

$$M_U(\alpha)(h_i) = \alpha \cdot h_i = \sum_{j=1}^{n} m_{i,j} h_j$$

where $m_{i,j}$ are now integers.

In other words, $M_U(\alpha)$ is an $n \times n$ matrix over $\mathbb{Z}$.

As above, we see that α is an *eigenvalue* of this matrix when we consider it as a matrix with entries in K. It follows that α satisfies the characteristic polynomial $\det(XI_n - M_U(\alpha))$ of $M_U(\alpha)$. Since $M_U(\alpha)$ has integer coefficients, this is a monic polynomial with integer coefficients. Thus, we conclude:

> An element α in K if integral if and only if there is a non-zero finitely generated subgroup $H \subset K$ such that $\alpha \cdot H \subset H$.

As seen above, such a finitely generated subgroup H is a free Abelian group containing a $\mathbb{Q}$-basis of the vector space $\mathbb{Q} \cdot H \subset K$.

The ring of algebraic integers

Given α and β in K which are integral. As seen above, we have a finitely generated subgroup $G \subset K$ (respectively $H \subset K$) such that $\alpha \cdot G \subset G$ (respectively $\beta \cdot H \subset H$).

Exercise: If $G \cdot H$ denotes the collection of (finite) sums of products of the form $g \cdot h$ with $g \in G$ and $h \in H$, then show that $G \cdot H$ is a finitely generated subgroup of K.

Note that $\alpha \cdot G \cdot H \subset G \cdot H$ and $\beta \cdot G \cdot H = G \cdot \alpha \cdot H \subset G \cdot H$.

It follows that $(\alpha + \beta) \cdot (G \cdot H) \subset G \cdot H$ and $(\alpha \cdot \beta) \cdot (G \cdot H) \subset G \cdot H$.

Hence, $\alpha + \beta$ and $\alpha \cdot \beta$ are integral as well.

> The sum and product of a pair of integral elements of K are themselves integral elements of K.

We use $\mathcal{O}_K$ to denote the *subset* of K that consists of algebraic integers. By what we have proved above, we see that $\mathcal{O}_K$ is closed under addition and multiplication. It is clear that 1 and 0 are algebraic integers. If α is an algebraic integer, then it is clear that $-\alpha$ is an algebraic integer.

> The collection $\mathcal{O}_K$ of algebraic integers in K forms a subring of K.

Matrix representation

Now consider the case where K is itself a finite dimensional vector space over $\mathbb{Q}$. This means that every element of K is algebraic.

Given a $\mathbb{Q}$-basis $S = \{v_1, \dots, v_n\}$ of K we have seen that there is a ring homomorphism $M_S : K \to M_n(\mathbb{Q})$ given by $\alpha \mapsto M_S(\alpha)$; the matrix $M_S(\alpha)$ has entries $a_{i,j}$ such that $\alpha \cdot v_i = \sum_{j=1}^{n} a_{i,j} v_j$.

Exercise: Note that for $\alpha \neq 0$, the matrix $M_S(\alpha)$ is invertible. (Multiplication by $\alpha \neq 0$ is a bijection from K to itself.)

For each i, the matrix $M_S(v_i)$ has entries in $\mathbb{Q}$. Since S is finite, there is a non-zero integer k such that, for each i, the matrix $M_S(kv_i)$ has entries in $\mathbb{Z}$.

Exercise: Check if $T = \{kv_1, \dots, kv_n\}$, then $M_T(\alpha) = M_S(\alpha)$ for all α in K.

> There is a basis $T = \{w_1, \dots, w_n\}$ of K over $\mathbb{Q}$ such that, for each i, the matrix $M_T(w_i)$ has entries in $\mathbb{Z}$.

Note that a general element α of K is a *rational* linear combination of w_i and thus $M_T(\alpha)$ is a matrix with entries in $\mathbb{Q}$.

Order. For a basis T of the vector space K over $\mathbb{Q}$, we define R_T to be the subset of K that consists of those α for which $M_T(\alpha)$ has entries in $\mathbb{Z}$. Since $M_n(\mathbb{Z})$ is a subring of $M_n(\mathbb{Q})$ it follows that R_T is a subring of K. Moreover, since R_T is a subgroup of the free group $M_n(\mathbb{Z})$, it is also a free group. In other words, there is a set $U = \{u_1, \dots, u_m\}$ of elements of R_T such that $R_T = \oplus_{k=1}^{m} \mathbb{Z} u_k$.

As seen above, we may assume that R_T contains T. Hence, every element of T is a *unique* $\mathbb{Z}$-linear combination of the elements of U. It follows that $m = n$ and U is *itself* a basis of K.

If $M_U(\alpha)$ is an integer matrix then $\alpha \cdot R_T \subset R_T$ so that $\alpha = \alpha \cdot 1$ lies in R_T. It follows easily that $R_T = R_U$.

In summary, we have shown that if T is a basis of K over $\mathbb{Q}$, the ring R_T of elements α in K such that $M_T(\alpha)$ is an integer matrix is of the form $R_T = R_U$, where U is a basis of K over $\mathbb{Q}$.

> A subring R of K which is the $\mathbb{Z}$-linear span of a basis of K over $\mathbb{Q}$ is called an *order* in K.

By what has been proved above we see that elements of an order are integral.

Trace, Norm and characteristic polynomial. Given bases $S = \{v_1, \ldots, v_n\}$ and $T = \{w_1, \ldots, w_n\}$ of K over $\mathbb{Q}$, let G be the invertible matrix such that $w_i = \sum_{j=1}^{n} G_{i,j} v_j$. It follows that

$$\alpha \cdot w_i = \sum_{j=1}^{n} G_{i,j} \alpha \cdot v_j \qquad = \sum_{j=1}^{n} \sum_{k=1}^{n} G_{i,j} M_S(\alpha)_{j,k} v_k$$
$$\alpha \cdot w_i = \sum_{j=1}^{n} M_T(\alpha)_{i,j} w_j \qquad = \sum_{j=1}^{n} \sum_{k=1}^{n} M_T(\alpha)_{i,j} G_{j,k} v_k$$

Hence $G \cdot M_S(\alpha) = M_T(\alpha) \cdot G$.

It follows that $\mathrm{Trace}(M_S(\alpha))$ and $\det(M_S(\alpha))$ are independent of the choice of basis. So, we define

$$\mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = \mathrm{Trace}(M_S(\alpha))$$
$$\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = \det(M_S(\alpha))$$

Similarly, the characteristic polynomial $\mathrm{ch}_{K/\mathbb{Q}}(\alpha)(X) = \det(X - M_S(\alpha))$ is independent of the choice of basis; it is a monic polynomial with coefficients in the subring of $\mathbb{Q}$ generated by the entries of $M_S(\alpha)$.

By the Cayley-Hamilton theorem, we see that α satisfies the polynomial $\mathrm{ch}_{K/\mathbb{Q}}(\alpha)$.

If there is *some* basis S of K in which $M_S(\alpha)$ has integer coefficients, then

- $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha)$ and $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$ are integers,
- $\mathrm{ch}_{K/\mathbb{Q}}(\alpha)$ has integer coefficients, and
- α is an algebraic integer.

Inner product. Given $\alpha \neq 0$, we see that $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha \cdot \alpha^{-1}) = n \neq 0$. It follows that the pairing

$$(,) : K \times K \to \mathbb{Q} \text{ given by } (\alpha, \beta) = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha \cdot \beta)$$

is a non-degenerate bilinear symmetric pairing on K as a vector space over $\mathbb{Q}$.

Thus, for $S = \{v_1, \ldots, v_n\}$ a basis of K, the matrix B_S with entries $(B_S)_{i,j} = (v_i, v_j)$ is an invertible symmetric matrix over $\mathbb{Q}$.

The Dual Group of an order R. Given an order R in K, we define the *dual* as the collection R^* of all $\alpha \in K$ such that $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha R) \subset \mathbb{Z}$.

Note that R is contained in R^*.

Now, suppose that $S = \{v_1, \ldots, v_n\}$ is a $\mathbb{Z}$-basis of the order R. Since R is a ring, $v_i \cdot v_j$ are in R. Hence, they are algebraic integers. Thus the matrix B_S has integer entries.

If $\alpha \in R^*$ is expressed as $\alpha = \sum_{i=1}^n a_i v_i$, then we have

$$(\alpha, v_j) = \sum_{i=1}^n a_i (B_S)_{i,j}$$

We see that R^* can be identified with the collection of vectors $\mathbf{a} = (a_i)$ in $\mathbb{Q}^n$ such that $B_S\mathbf{a}$ is a vector with integer entries. In other words, it can be identified with $(B_S)^{-1}\mathbb{Z}^n$ with respect to the basis S.

In particular, we note that R^* is *also* the $\mathbb{Z}$-linear span of a basis and is contained in $(1/\det(B_S))R$.

> If $S = \{v_1, \ldots, v_n\}$ is a basis of on order R then R^* contains R and is contained in the subgroup $(1/\det(B_S))R$.

The ring of algebraic integers in K

We have seen that, if R is an order in K, then it consists of algebraic integers, so $R \subset \mathcal{O}_K$.

We have also seen that if $\alpha \in K$ is an algebraic integer then $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha)$ lies in $\mathbb{Z}$.

So, if α is an element of $\mathcal{O}_K$, then $\alpha \cdot R \subset \mathcal{O}_K$ implies that $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha \cdot R) \subset \mathbb{Z}$. In other words, $\alpha \in R^*$.

This means that $\mathcal{O}_K \subset R^*$; so $\mathcal{O}_K$ is "trapped" between the free Abelian groups R and $R^* \subset (1/N)R$ for some positive integer N. Thus, $\mathcal{O}_K$ is itself of the form $\oplus_{i=1}^n \mathbb{Z} \cdot \alpha_i$ for some algebraic integers $\alpha_1, \ldots, \alpha_n$.

Putting $S = \{\alpha_1, \ldots, \alpha_n\}$, we see that $\mathcal{O}_K = R_S$ is an order. It follows easily that $\mathcal{O}_K$ is a *maximal* order in K, in the sense that it is an order which contains all other orders.

We note that $\det(B_S)$ is *independent* of the choice of $\mathbb{Z}$-basis of $\mathcal{O}_K$ since a change of basis will lead to multiplication by the *square* of the associated change of variable matrix. Thus, it is an "invariant" associated with the field K; this invariant is called the *discriminant of the field* K and is denoted as D_K.

Let the dual of the order $\mathcal{O}_K$ be $(\mathcal{O}_K)^*$. Note that it contains $\mathcal{O}_K$ and is contained in $(1/D_K)\mathcal{O}_K$. We define the *different* δ_K to be the collection of α in K such that $\alpha \cdot (\mathcal{O}_K)^* \subset \mathcal{O}_K$.

Hermite-Minkowski Theorem

We see easily that $D_{\mathbb{Q}} = 1$.

Hermite proved that given a degree n and a positive integer N, there are at most finitely many number fields of degree n over $\mathbb{Q}$ whose discriminant is at most N.

Minkowski proved a bound which shows that the degree of the field extension is also bounded in terms of the discriminant. This implies that the *only* number field K with $D_K = 1$ is $K = \mathbb{Q}$. It also improves Hermite's theorem to show that given a positive integer N, there are at most finitely many number fields whose discriminant is at most N.